Report 1: Are Headtracking Measures an Indicator of Depressive Symptoms?

Preliminary Data Visualization, Descriptive Statistics, and Inferential Analysis

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1 Introduction

Depression is a leading cause of disability worldwide, yet objective behavioral markers remain limited. While self-report instruments such as the Patient Health Questionnaire (PHQ-9; Kroenke et al. 2001) are widely used for screening, they are inherently subjective. An emerging line of research investigates *psychomotor retardation*—the slowing of physical movement and thought—as an observable correlate of depression (Bennabi et al., 2013). The anhedonia component of depression, in particular, may manifest as reduced exploratory behavior when individuals are placed in stimulating environments.

Recent work by Srivastava et al. (2025) demonstrated that headtracking data collected during 360° virtual reality (VR) video viewing could distinguish individuals based on depression severity. Using ANCOVA to control for anxiety (GAD-7 and STAI-T), they found that *scanning speed*—the rate of angular head movement across the visual field—was significantly lower in the moderate-severe depression group compared to mild and minimal groups ($p < .001$, $\eta^2 = .295$, Cohen's $d = 1.64$). The present study is a **replication** of this work, conducted with college students in India using a Meta Quest 3 VR headset.

Participants viewed five 360° videos spanning different emotional contexts (abandoned buildings, a beach, a campus, a horror scene, and surfing) and completed standardized measures of depression (PHQ-9), anxiety (GAD-7; Spitzer et al. 2006; STAI-T; Spielberger et al. 1983), current mood (PANAS-SF; Watson et al. 1988), presence, and simulator sickness (VRNQ-VRISE; Kourtesis et al. 2019).

Since depression and anxiety are highly covariant constructs (Kalin, 2020), any analysis of depression-related differences must account for this overlap. This report presents preliminary descriptive statistics, visualizations, and initial inferential tests to characterize the dataset and set the stage for more advanced analyses in Report 2.

2 Methods

2.1 Participants

Forty college students (36 male, 4 female; age $M = 22.8$, $SD = 1.8$, range 19–27) participated. Sixteen (40%) reported prior VR experience. Participants were recruited via convenience sampling.

2.2 Materials

Five 360° videos were used:

- **V1**: Abandoned Buildings (60s) — eerie, low valence
- **V2**: Evening at the Beach (60s) — pleasant, calm
- **V3**: Campus (60s) — neutral, familiar
- **V4**: The Nun Horror (≈3 min) — high arousal, negative
- **V5**: Tahiti Surf (≈3 min) — high arousal, positive

Videos were viewed on a Meta Quest 3 headset. Surveys were administered via PsyToolKit.

2.3 Measures

1. **Headtracking**: Pitch (X), Yaw (Y), and Roll (Z) rotation changes and speeds recorded at ≈70 Hz. Summary measures computed per participant per video: mean rotation speed (total and per axis), standard deviation, median speed, mean absolute rotation change, range of angular exploration, and peak speed.
2. **PHQ-9**: 9-item depression screener (0–27). Clinical cutoff of ≥ 10 used for moderate depression (Kroenke et al., 2001).
3. **GAD-7**: 7-item generalized anxiety measure (0–21) (Spitzer et al., 2006).
4. **STAI-T**: 20-item trait anxiety inventory (20–80) (Spielberger et al., 1983).
5. **PANAS-SF**: 20-item affect schedule measuring positive and negative affect pre- and post-VR (Watson et al., 1988).
6. **VRNQ-VRISE**: 5-item simulator sickness subscale (5–35) (Kourtesis et al., 2019).
7. **Emotion ratings**: Valence (1–9, unpleasant–pleasant) and Arousal (1–9, calm–excited) per video.
8. **Presence**: 5-item immersion measure (5–35) per video.

2.4 Depression Group Classification

Participants were classified as *Depressed* (PHQ-9 ≥ 10; $n = 8$) or *Non-Depressed* (PHQ-9 < 10; $n = 32$) using the established clinical cutoff for moderate depression (Kroenke et al., 2001). The original study by Srivastava et al. (2025) used three depression groups (minimal, mild, moderate-severe) with ≥ 10 marking the moderate-severe threshold; we adopt the same cutoff for a binary split given our smaller sample.

2.5 Statistical Approach

- **Descriptive statistics**: Means, standard deviations, medians, and ranges for all key variables.
- **Normality testing**: Shapiro-Wilk tests to assess distributional assumptions.
- **Correlations**: Pearson correlations for bivariate relationships between clinical measures and headtracking variables.
- **Group comparisons**: Welch's independent-samples t-tests comparing headtracking measures across depression groups. Although Shapiro-Wilk tests indicated non-normality for some clinical measures, t-tests are robust to moderate departures from normality, especially with $N = 40$ (Central Limit Theorem). Welch's correction was applied to avoid the equal-variance assumption. Effect sizes are reported as Cohen's d.
- **Multiple comparison correction**: Bonferroni correction applied to the 25 simultaneous t-tests ($\alpha_{\text{corrected}} = 0.05/25 = 0.002$).
- **Pre-post mood changes**: Paired t-tests for within-subject PANAS changes.
- **Outlier detection**: IQR method for clinical scores; z-score method ($|z| > 3$) for headtracking data.

All analyses were conducted in Python 3.12 using `pandas`, `scipy`, `seaborn`, and `matplotlib`. Significance was set at $\alpha = 0.05$ unless otherwise noted.

3 Results

3.1 Demographics and Sample Characteristics

The sample was predominantly male (90%) with a narrow age range (19–27 years). Figure 1 shows the demographic distributions.

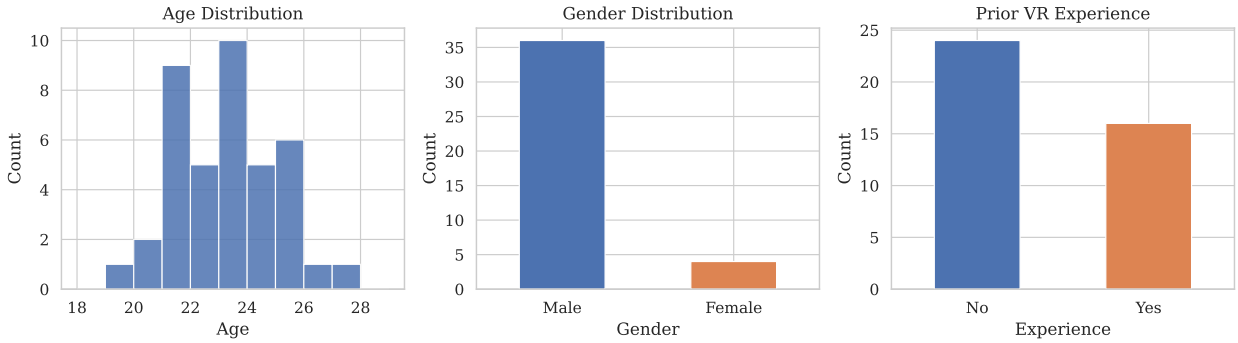


Figure 1: Distributions of age, gender, and prior VR experience across the 40 participants.

3.2 Clinical Measures

PHQ-9 scores ranged from 0 to 18 ($M = 6.03$, $SD = 4.63$, $Mdn = 4.5$), with 50% of participants in the minimal range (0–4), 30% mild (5–9), 12.5% moderate (10–14), and 7.5% moderately severe (15–19). Eight participants (20%) met the threshold for moderate depression (PHQ-9 ≥ 10). GAD-7 scores ranged from 0 to 18 ($M = 5.00$, $SD = 4.31$). STAI-T scores ranged from 21 to 73 ($M = 45.00$, $SD = 14.63$).

Shapiro-Wilk tests indicated that both PHQ-9 ($W = 0.897$, $p = .002$) and GAD-7 ($W = 0.883$, $p < .001$) distributions deviated significantly from normality, with a positive skew reflecting the predominantly non-clinical sample. STAI-T was marginally non-normal ($W = 0.944$, $p = .045$).

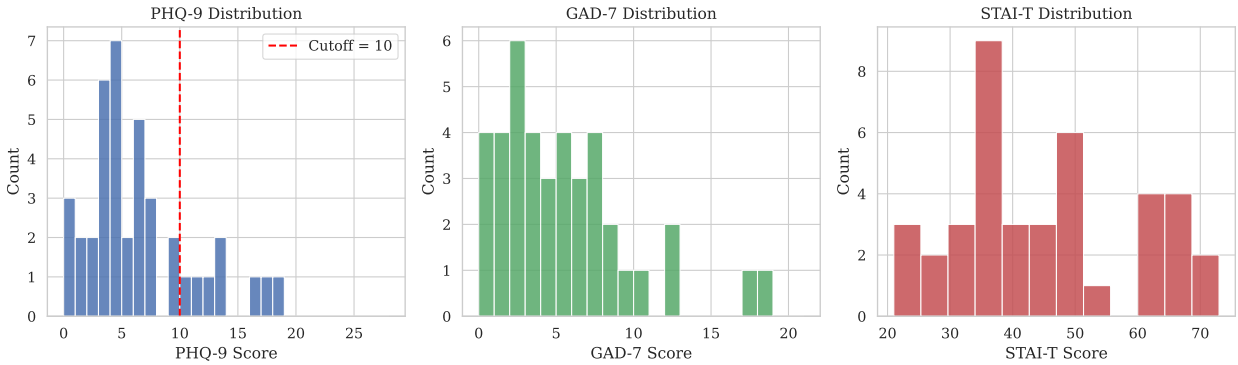


Figure 2: Distributions of PHQ-9, GAD-7, and STAI-T scores. The dashed red line on PHQ-9 indicates the clinical cutoff (≥ 10) for moderate depression.

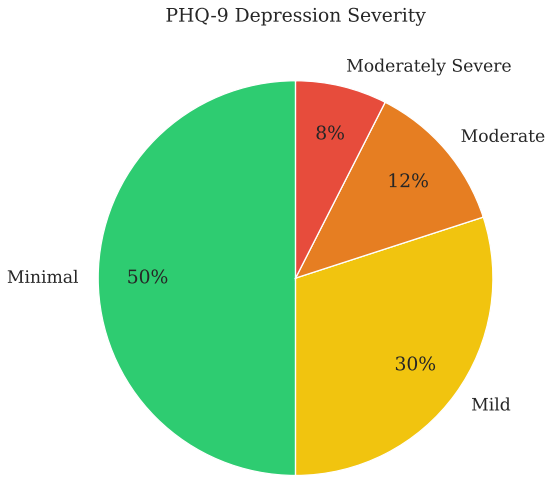
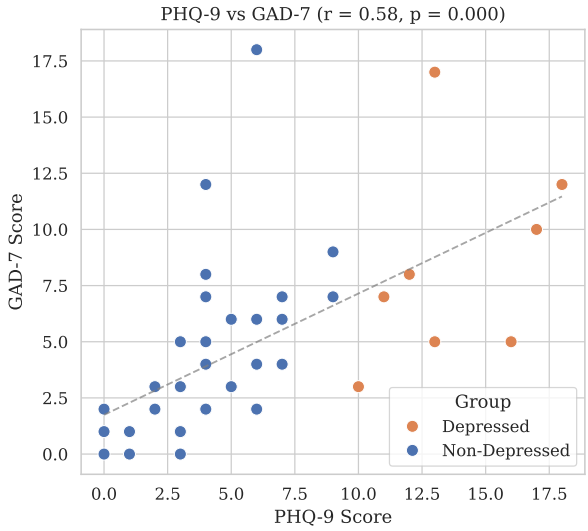


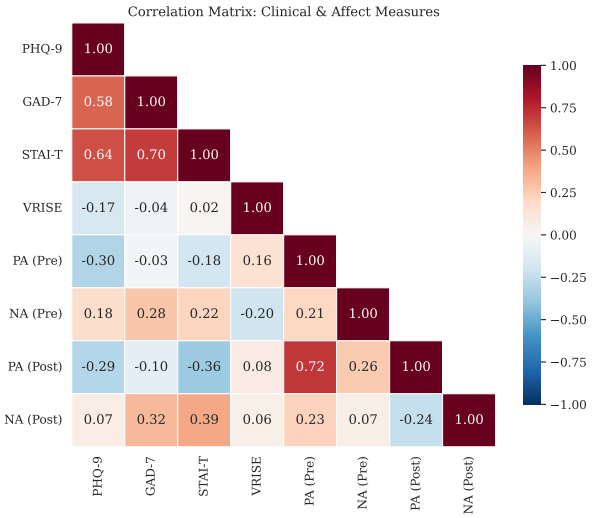
Figure 3: PHQ-9 depression severity classification of participants.

3.3 Depression–Anxiety Covariance

PHQ-9 and GAD-7 were strongly correlated (Pearson $r = .579$, $p < .001$), consistent with the well-documented comorbidity of depression and anxiety (Kalin, 2020). PHQ-9 and STAI-T were also strongly correlated ($r = .642$, $p < .001$). These moderate-to-strong correlations underscore the need to control for anxiety when isolating depression-specific effects on head-tracking behavior.



(a) PHQ-9 vs. GAD-7 scatter plot.



(b) Correlation heatmap of clinical and affect measures.

Figure 4: (a) Scatterplot showing the strong positive relationship between PHQ-9 and GAD-7 scores. (b) Correlation matrix showing inter-relationships among all clinical and affect measures.

3.4 Video-Level Emotional and Presence Ratings

The five videos elicited distinct emotional profiles (Table 1). V5 (Tahiti Surf) received the highest valence ($M = 7.12$) and arousal ($M = 6.50$), while V4 (Horror) had the lowest valence ($M = 3.67$) but high arousal ($M = 5.95$). V1 (Abandoned Buildings) scored highest on presence ($M = 29.60$), while V4 scored lowest ($M = 19.85$).

Table 1: Mean (SD) valence, arousal, and presence scores by video.

Video	**Valence**	**Arousal**	**Presence**
V1: Abandoned Buildings	5.20 (1.67)	4.60 (2.22)	29.60 (4.41)
V2: Beach	6.53 (1.75)	4.65 (2.03)	27.25 (5.31)
V3: Campus	6.15 (1.61)	4.33 (2.16)	26.95 (4.52)
V4: Horror (Nun)	3.67 (1.85)	5.95 (1.68)	19.85 (5.44)
V5: Tahiti Surf	7.12 (1.68)	6.50 (1.69)	28.02 (4.79)

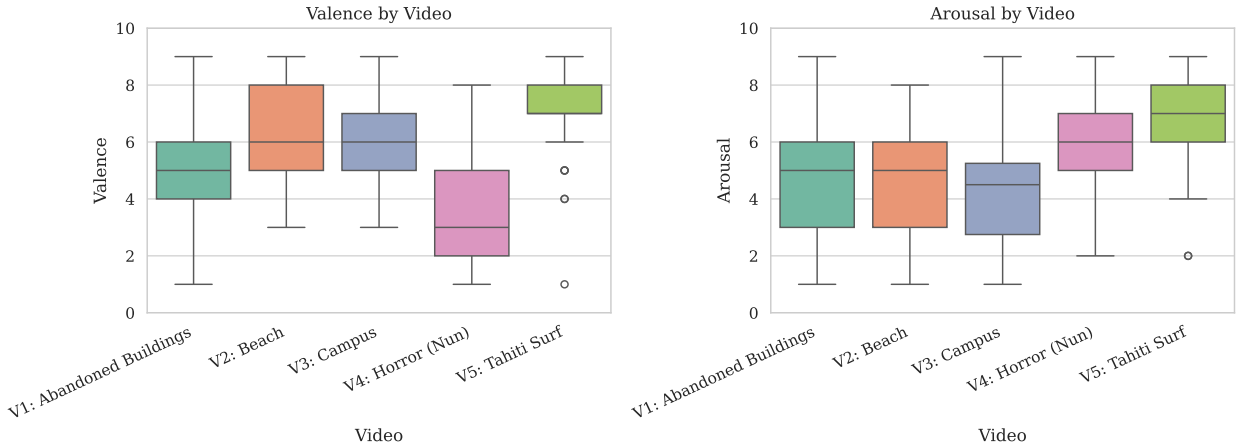


Figure 5: Box plots of valence and arousal ratings across the five videos.

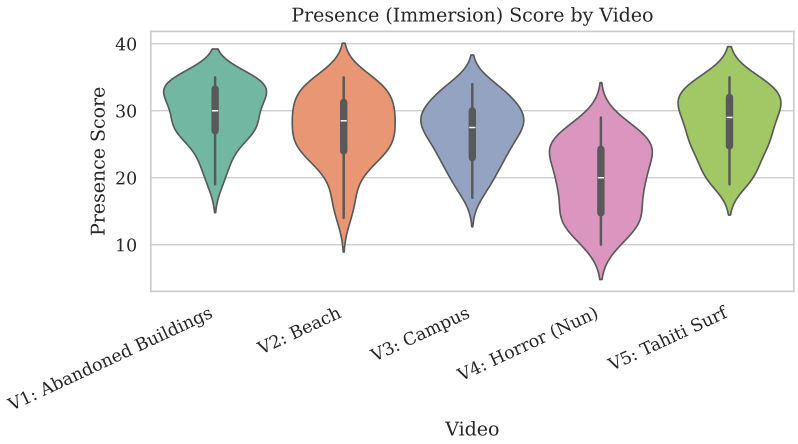


Figure 6: Violin plots of presence (immersion) scores by video. V4 (Horror) had notably lower presence, possibly due to lower visual quality.

3.5 Headtracking Measures

Descriptive analysis revealed that mean rotation speed varied across videos. V1 (Abandoned Buildings) elicited the highest mean speed ($M = 39.08°/s$), consistent with the exploratory nature of the environment, while V4 (Horror) had the lowest ($M = 24.17°/s$), possibly because the horror narrative directed gaze toward specific focal points. Formal testing of speed differences across videos (one-way ANOVA) is deferred to Report 2.

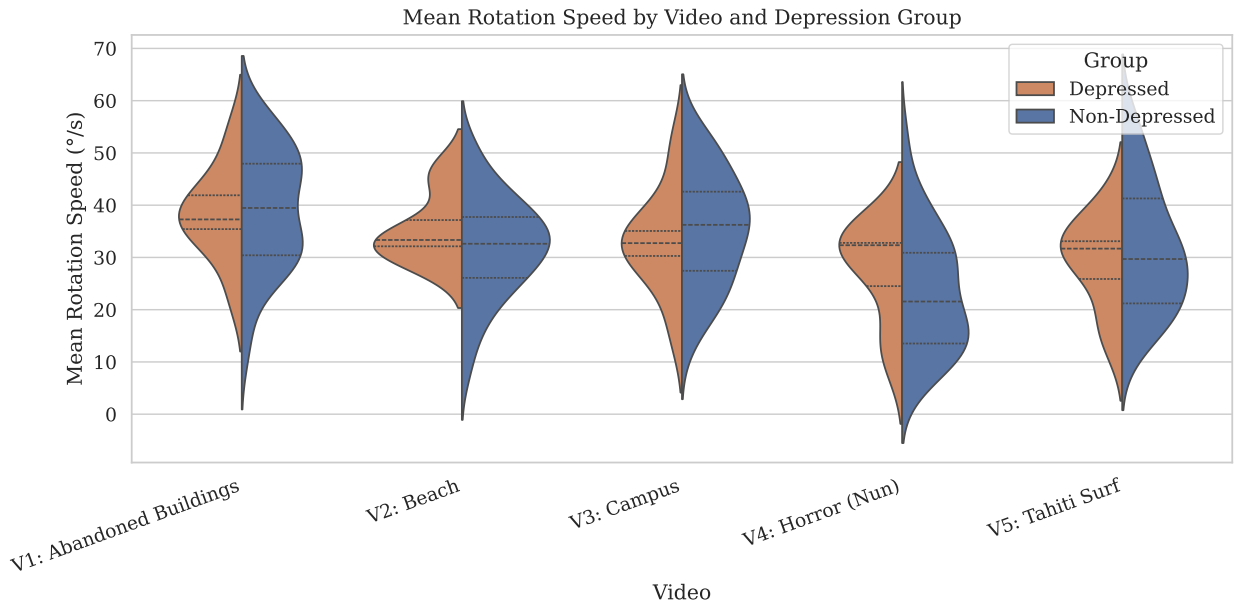


Figure 7: Violin plots of mean rotation speed by video and depression group. Split violins show the distribution for Non-Depressed (blue) and Depressed (orange) groups.

3.6 Headtracking by Depression Group

Welch's independent-samples t-tests comparing depressed and non-depressed groups on headtracking measures yielded largely non-significant results (Table 2). At the uncorrected $\alpha = .05$ level, only one of 25 comparisons reached significance: **V4 (Horror)** total angular range ($t = 2.21$, $p = .048$, Cohen's $d = 0.83$), where the depressed group explored a *greater* angular range. After Bonferroni correction ($\alpha_{\text{corrected}} = .002$ for 25 comparisons), **no comparisons remained significant**. This finding runs counter to the psychomotor retardation hypothesis and is likely a Type I error.

Pearson correlations between PHQ-9 scores and participant-averaged headtracking measures were near zero and non-significant: mean total speed ($r = -.060$, $p = .712$), mean yaw speed ($r = -.081$, $p = .617$), and SD of total speed ($r = -.158$, $p = .329$).

Table 2: Selected Welch's t-test results for mean total speed and total angular range by depression group per video. Cohen's d quantifies effect size. No comparison survived Bonferroni correction ($\alpha_{\text{corrected}} = .002$).

Video	**Mean Speed (Total)**			**Total Angular Range**		
	t	p	d	t	p	d
V1: Abandoned Buildings	−0.24	.816	−0.08	−0.22	.831	−0.07
V2: Beach	1.22	.239	0.38	1.76	.104	0.65
V3: Campus	−0.64	.532	−0.24	0.16	.878	0.05
V4: Horror (Nun)	1.07	.303	0.37	2.21	.048	0.83
V5: Tahiti Surf	−0.66	.523	−0.21	−0.36	.727	−0.16

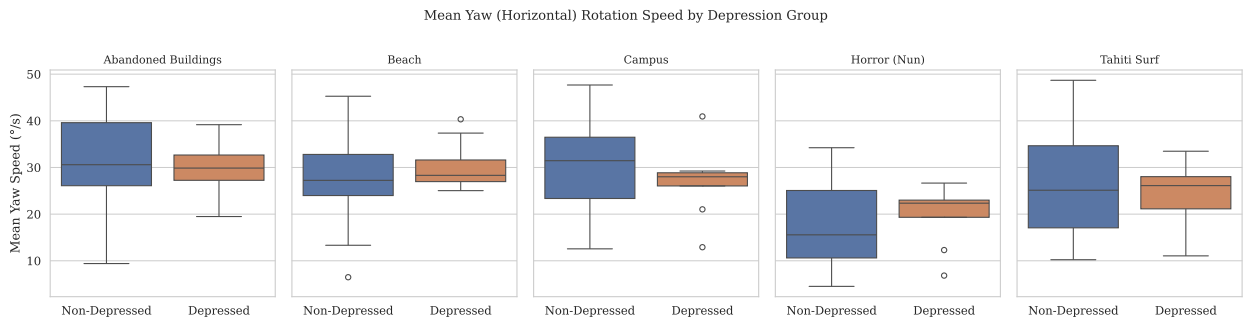


Figure 8: Mean yaw (horizontal) rotation speed by depression group for each video.

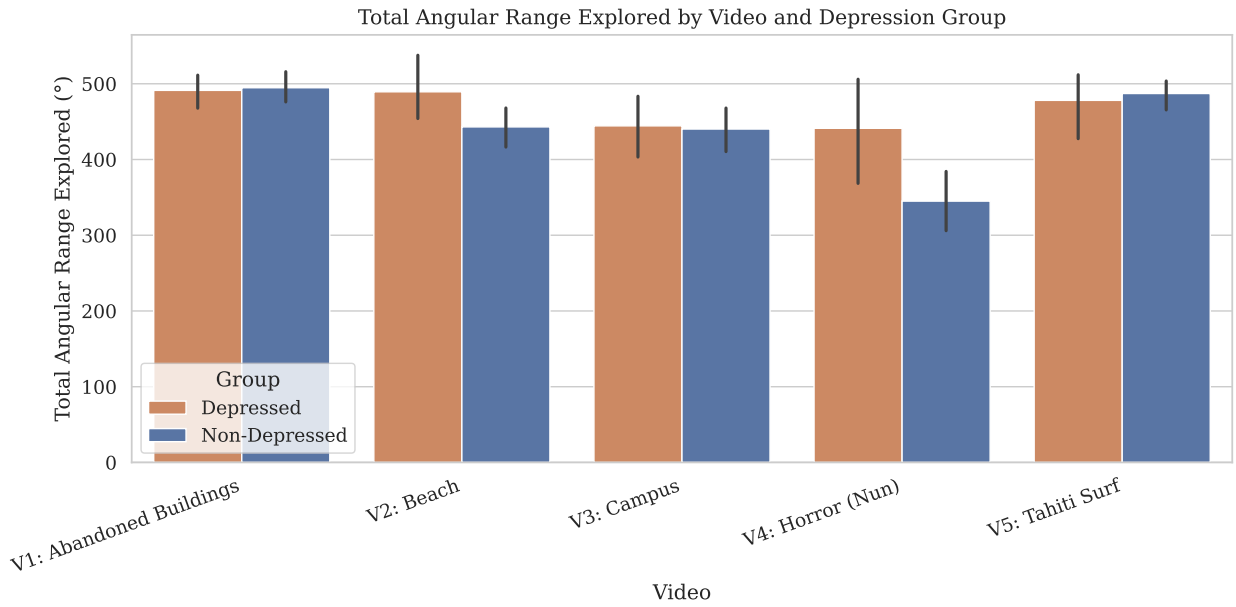


Figure 9: Total angular range explored by video and depression group. Error bars represent 95% confidence intervals.

3.7 PANAS: Pre- and Post-VR Mood

Positive Affect decreased significantly from pre-VR ($M = 34.52$) to post-VR ($M = 32.15$; paired $t(39) = 2.09$, $p = .043$, Cohen's $d = -0.33$). Negative Affect showed a non-significant increase ($M = 12.70 \rightarrow 13.95$; $t(39) = -1.06$, $p = .294$, $d = 0.17$).

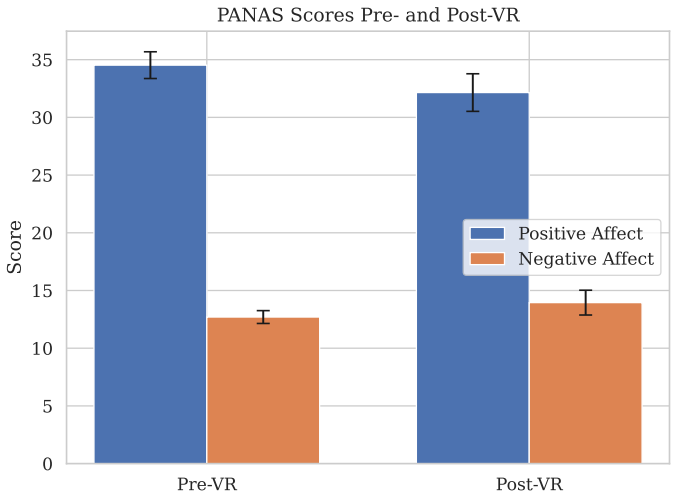


Figure 10: PANAS positive and negative affect scores before and after VR exposure. Error bars represent standard errors.

3.8 Outlier Analysis

Using the IQR method, 3 PHQ-9 outliers (scores > 14.2) and 2 GAD-7 outliers (scores > 14.5) were identified. However, these represent genuine clinical variability (moderately severe depression/anxiety) and were retained. No headtracking speed outliers exceeded $|z| > 3$. One participant had a low VRISE score (17), suggesting possible simulator sickness; this participant was retained but flagged for sensitivity analysis.

4 Conclusion and Future Directions

4.1 Summary of Findings

1. The sample is predominantly non-clinical: 80% of participants scored below the moderate depression threshold (PHQ-9 < 10), with only 8 in the "depressed" group.
2. Depression and anxiety are strongly correlated (Pearson $r = .58$), confirming the need to account for this covariance in subsequent analyses.
3. Videos elicited distinct emotional profiles, and descriptive patterns suggest headtracking speed varies across videos (V1 highest at 39.08°/s, V4 lowest at 24.17°/s). Formal one-way ANOVA testing is deferred to Report 2.

4. Welch's t-tests showed **no significant differences** in mean headtracking speed between depressed and non-depressed groups for any video. After Bonferroni correction for 25 comparisons ($\alpha_{\text{corrected}} = .002$), no comparison reached significance. The original study's finding of significantly reduced scanning speed in the moderate-severe depression group (Srivastava et al., 2025) was **not replicated** in this preliminary analysis, likely due to limited statistical power.

5. One uncorrected-significant effect was found for angular range in V4 (Horror) ($t = 2.21$, $p = .048$, $d = 0.83$), but in the opposite direction to expectations and non-significant after Bonferroni correction—likely a Type I error.

6. Positive affect decreased significantly after VR exposure (paired t-test: $p = .043$, $d = -0.33$).

4.2 Limitations

The key limitation is the small and unbalanced sample ($n_{\text{dep}} = 8$ vs. $n_{\text{nondep}} = 32$), which severely limits statistical power. The predominantly male sample (90%) limits generalizability. Additionally, the current binary classification does not exploit the full range of PHQ-9 scores.

4.3 Plans for Report 2

1. Use PHQ-9 as a continuous predictor in regression/GLM models, rather than relying solely on categorical grouping, to exploit the full range of scores.

2. Include GAD-7 and STAI-T as covariates (ANCOVA or multiple regression) to disentangle depression from anxiety effects on headtracking, following the approach of citetsrivastava2025headscan.

3. Conduct repeated-measures analyses (mixed-effects models or repeated-measures ANOVA) to account for the within-subject structure across 5 videos.

4. Investigate video-type interactions: do different emotional contexts (positive, negative, neutral, horror) moderate the depression–headtracking relationship?

5. Perform mediation/moderation analysis with presence and simulator sickness scores.

6. Conduct effect-size-based power analysis to quantify the statistical power of the current sample and determine what effect sizes are detectable.

7. Compare results using alternative multiple-comparison corrections (Holm, Benjamini-Hochberg) alongside Bonferroni.

8. Conduct one-way ANOVA (F-test) to formally test whether mean headtracking speed differs significantly across the five videos, with η^2 as effect size.

References

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